Name: _____ Date: _____ Period: _____

HONORS CHEMISTRY • UNIT 6

Stoichiometry

Use a balanced equation like a recipe: predict how much you need, how much you make, what runs out first, and how well the lab went.

HOW TO USE THIS PACKET

Write in pencil. Every calculation needs a setup with units, an answer with the right number of sig figs, and the substance named. LAB CORNER boxes are things you would see at a lab bench. Extra practice and a review are at the back.

What we will learn

6.1	Mole ratios
6.2	Mass-to-mass stoichiometry
6.3	Liters and particles (gas volume at STP)
6.4	Limiting and excess reactants (BCA tables)
6.5	Theoretical, actual and percent yield

By the end of this unit I can...

- Write mole ratios from a balanced equation and use them to convert between moles of two substances.
- Solve mass-to-mass problems using a mole map and show every unit.
- Convert between liters of gas at STP, moles, grams and particles.
- Identify the limiting reactant and calculate product made and excess left over using a BCA table.
- Calculate theoretical yield and percent yield, and explain why actual yield is smaller.

Name: _____ Date: _____ Period: _____

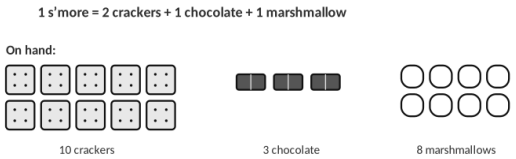
6.1 Mole Ratios *The coefficients of a balanced equation are a recipe*

BIG IDEA

A balanced equation tells you how many **particles** react, and since a mole is a counting number, the coefficients also give **moles**. The **mole ratio** is a conversion factor built from two coefficients. It is the only bridge between one substance and another.

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Stoichiometry	*What is it used for?*	
Mole ratio	*Where does it come from?*	



Analogy: the recipe 2 crackers : 1 chocolate : 1 marshmallow plays the role of the coefficients.

1. Use the s'more recipe. How many s'mores can you make from 6 chocolates if you have plenty of everything else? How many crackers does that use? Write the two ratios you used.

2. What do you need before you can write a mole ratio? Why must the equation be balanced first?

Write the mole ratios

Equation	Ratios (write each as a fraction)
$N_2 + 3H_2 \rightarrow 2NH_3$	N_2/H_2 = _____ N_2/NH_3 = _____ H_2/NH_3 = _____
$8H_2 + S_8 \rightarrow 8H_2S$	H_2/H_2S = _____ H_2/S_8 = _____ H_2S/S_8 = _____

3. Hydrogen gas reacts with oxygen gas to make water. (a) Write the balanced equation. (b) Write the ratios H_2/O_2, H_2/H_2O and O_2/H_2O. (c) How many moles of water form from 10 moles of hydrogen?

a. Balanced equation

Work:

Answer (units & sig figs): _____

b. Mole ratios

Work:

Answer (units & sig figs): _____

c. Moles of water

Work:

Answer (units & sig figs): _____

WORKED EXAMPLE — MOLE RATIO

How many moles of NH_3 form from 4.0 mol N_2? $N_2 + 3H_2 \rightarrow 2NH_3$

Work it out here:

Answer: _____

4. Use $N_2 + 3H_2 \rightarrow 2NH_3$.

a. If 10 moles of NH_3 are produced, how many moles of N_2 were needed?

Work:

Answer (units & sig figs): _____

b. If 3.00 moles of H_2 are used, how many moles of NH_3 are made?

Work:

Answer (units & sig figs): _____

c. If 0.600 moles of NH_3 are produced, how many moles of H_2 are required?

Work:

Answer (units & sig figs): _____

5. The reaction between hydrazine (N_2H_2) and hydrogen peroxide (H_2O_2) fuels rockets. The products are nitrogen gas and water. (a) Write the balanced equation. (b) How many moles of N_2 form from 37 moles of hydrazine? (c) If 23 moles of water form, how many moles of H_2O_2 were used?

a. Balanced equation

Work:

Answer (units & sig figs): _____

b. Moles of N_2

Work:

Answer (units & sig figs): _____

c. Moles of H_2O_2

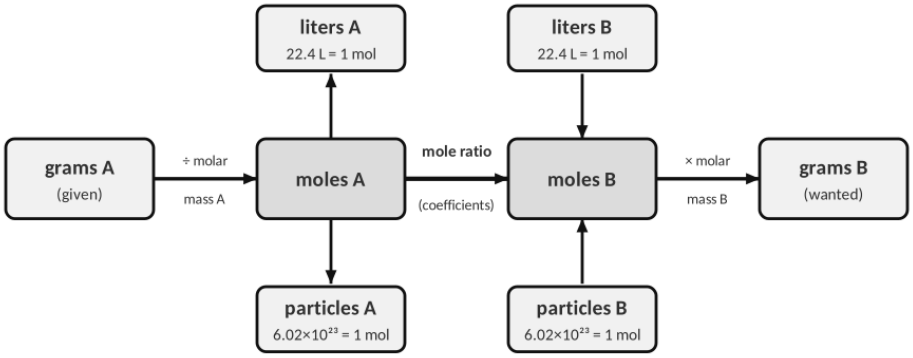
Work:

Answer (units & sig figs): _____

6.2 Mass-to-Mass Stoichiometry *Grams → moles → moles → grams*

BIG IDEA

You cannot compare grams of different substances directly, because a gram of one is not the same number of particles as a gram of another. Convert to **moles**, use the **mole ratio**, then convert back to **grams**.



The mole map (use the top and bottom boxes in 6.3). Molar masses come from the periodic table.

1. In your own words, why is the mole ratio step the only step that uses the balanced equation? What do the other two steps use?

SETUP RULES

1. Balance the equation. 2. Write the given quantity with its unit and substance. 3. Chain factors so that each unit cancels. 4. Round the final answer to the sig figs of the given data.

WORKED EXAMPLE — GRAMS TO GRAMS

How many grams of FeS form from 5.9 g of Fe? ($8Fe + S_8 \rightarrow 8FeS$)

Work it out here:

Answer: _____

2. Use the equation $Fe + S_8 \rightarrow FeS$. (a) Balance it.

a. Balance the equation

Work:

Answer (units & sig figs): _____

b. How many moles of FeS are produced when 2.4 g of S_8 react?

Work:

Answer (units & sig figs): _____

c. How many grams of iron are used if 11.9 g of sulfur (S_8) are used?

Work:

Answer (units & sig figs): _____

3. Use the equation $Fe_2O_3 + C \rightarrow Fe + CO_2$ (the reaction in a blast furnace).

a. Balance the equation

Work:

Answer (units & sig figs): _____

b. How many grams of iron form if 39.64 g of carbon are used?

Work:

Answer (units & sig figs): _____

c. If 63.12 g of iron(III) oxide are used, how many grams of carbon dioxide form?

Work:

Answer (units & sig figs): _____

4. A student solved "grams of Fe_2O_3 → grams of CO_2" by multiplying by 159.7 g/mol and then by 44.01 g/mol, skipping the mole ratio. Describe what is wrong and what the student's answer would represent.

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5. To convert grams of A to grams of B you need…

A) molar masses only B) molar masses and the mole ratio C) the mole ratio only D) 22.4 L/mol

6.3 Liters and Particles *Gas volume at STP and counting atoms*

BIG IDEA

At **STP** (0 °C and 1 atm), **1 mole of any gas occupies 22.4 L**. This lets us convert between liters of a gas and moles, and so to any other substance. One mole is also 6.02×10^{23} particles.

Name: _____ Date: _____ Period: _____

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
STP	*Two conditions and a number*	
Molar volume	*Units?*	

THINK OF IT LIKE… A DOZEN IS JUST A NUMBER

A dozen is 12 things, whether they are eggs or elephants. A mole is 6.02×10^{23} things, and at STP that many gas molecules take up 22.4 L whether they are H_2 or Cl_2. Gases are mostly empty space, so the size of the molecule hardly matters.

1. Explain why 1 mole of H_2 and 1 mole of Cl_2 occupy the same volume at STP even though a Cl_2 molecule is much heavier.

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2. Hydrogen gas and oxygen gas combine to form water. Assume STP.

a. How many liters of H_2 are needed to make 3 moles of water?

Work:

Answer (units & sig figs): _____

b. How many grams of water form from 30.2 L of hydrogen?

Work:

Answer (units & sig figs): _____

c. How many grams of water form from 30.2 L of oxygen?

Work:

Answer (units & sig figs): _____

d. When 120.3 g of water form, how many liters of hydrogen were used?

Work:

Answer (units & sig figs): _____

Name: _____ Date: _____ Period: _____

LAB CORNER • Sodium azide in a car air bag

A crash sensor sets off the reaction $NaN_3(s) \rightarrow Na(l) + N_2(g)$. The nitrogen gas inflates the bag in about 0.03 seconds.

[IMAGE GOES HERE]

Photo or diagram of an air bag inflator (add your own)

3. Balance the equation, then solve (STP).

a. Balance the equation

Work:

Answer (units & sig figs): _____

b. How many liters of N_2 form from 15.62 g of NaN_3?

Work:

Answer (units & sig figs): _____

c. How many grams of NaN_3 are needed to produce 200.0 L of N_2?

Work:

Answer (units & sig figs): _____

d. If 31.4 g of sodium are produced, how many liters of N_2 are produced?

Work:

Answer (units & sig figs): _____

4. How many molecules of water form from 5.0 g of hydrogen gas? ($2H_2 + O_2 \rightarrow 2H_2O$; 1 mol = 6.02×10^{23} molecules)

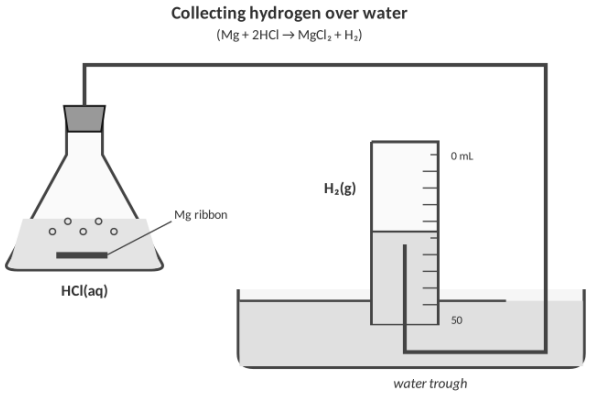
Work:

Answer (units & sig figs): _____

Name: _____ Date: _____ Period: _____

LAB CORNER • Hydrogen gas collected over water

A 0.150 g strip of magnesium reacts completely with excess hydrochloric acid. The hydrogen is collected in an inverted graduated cylinder.



5. (a) Write the balanced equation. (b) Calculate the theoretical volume of H_2 at STP in mL. (c) The measured volume was 128 mL. Name one reason it differs.

a. Equation

Work:

Answer (units & sig figs): _____

b. Theoretical volume

Work:

Answer (units & sig figs): _____

c. Reason

Work:

Answer (units & sig figs): _____

6.4 Limiting and Excess Reactants *Which ingredient runs out first?*

BIG IDEA

Reactants are almost never mixed in perfect recipe amounts. The **limiting reactant** is the one that runs out first and so decides how much product forms. The **excess reactant** is what is left over when the reaction stops.

Name: _____ Date: _____ Period: _____

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Limiting reactant	*What does it limit?*	
Excess reactant	*What happens to it?*	



Recipe: 2 crackers + 1 chocolate + 1 marshmallow → 1 s'more.

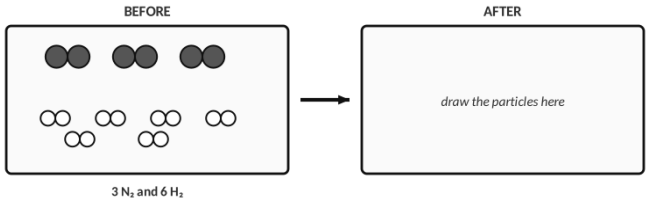
1. Using the s'more recipe and the items shown: (a) How many s'mores can you make? (b) What is the limiting "reactant"? (c) What is left over, and how many of each?

Work:

BCA TABLE STEPS (MOLES!)

1. Balance the equation. Convert every given amount to **moles**.
2. Divide each by its coefficient; the **smallest** value is the limiting reactant.
3. **Before** row = starting moles. **Change** row = the limiting reactant goes to 0; use the mole ratio for the rest (reactants −, products +).
4. **After** row = Before + Change. Convert the After amounts to the unit the question asks for.

Particle picture: $N_2 + 3H_2 \rightarrow 2NH_3$



2. Which reactant is limiting? Draw the particles after the reaction in the AFTER box, then say how many N_2 molecules are left.

WORKED EXAMPLE — BCA TABLE

1.5 mol CH_4 react with 2.5 mol Cl_2: $CH_4 + 2Cl_2 \rightarrow CH_2Cl_2 + 2HCl$. Find the moles of each product and the moles of excess reactant left.

Work it out here:

(mol)	CH_4	$2Cl_2$	CH_2Cl_2	2HCl
Before				
Change				
After				

Divide by coefficients and answer:

3. 15.0 g of N_2 gas reacts with 4.50 g of H_2 gas: $N_2 + 3H_2 \rightarrow 2NH_3$. Use a BCA table to find the mass of NH_3 made and the mass of the excess reactant left.

(mol)	N_2	$3H_2$	$2NH_3$
Before			
Change			
After			

Work and answers:

4. 17.5 L of NH_3 gas react with 12.5 L of O_2 gas at STP: $4NH_3$ (g) + $5O_2$ (g) → 4NO (g) + $6H_2O$ (g). Use a BCA table to find the mass of each product and the liters of excess reactant left.

(mol)	$4NH_3$	$5O_2$	4NO	$6H_2O$
Before				
Change				
After				

Work and answers:

5. 7.50 g of solid sodium react with 18.5 L of Cl_2 gas at STP: 2Na (s) + Cl_2 (g) → 2NaCl (s). Find the mass of NaCl made and the mass of the excess reactant left.

(mol)	2Na	Cl_2	2NaCl
Before			
Change			
After			

Work and answers:

6. A classmate says, "The limiting reactant is the one with the smaller mass." Use any example above to show why this is not reliable.

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7. In a BCA table, the limiting reactant's "After" value is always…

A) equal to its Before value B) zero C) equal to the product D) negative

6.5 Theoretical, Actual and Percent Yield *How well did the lab go?*

BIG IDEA

The **theoretical yield** is the most product the equation predicts (from the limiting reactant). The **actual yield** is what you really collect in the lab. **Percent yield = actual ÷ theoretical × 100**, and it can never honestly be more than 100 %.

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Theoretical yield	*Where does it come from?*	
Actual yield	*How is it found?*	
Percent yield	*Formula?*	

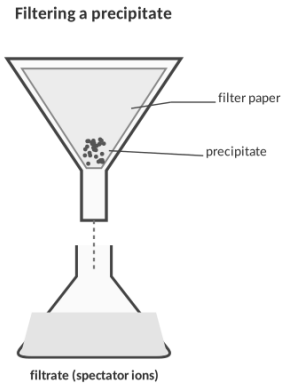
THINK OF IT LIKE… FREE THROWS

A player who shoots 10 free throws could, in theory, make all 10 (theoretical yield). Making 8 is the actual yield, so the percent yield is 80 %. Nobody can make 12 out of 10, so a percent yield over 100 % means something is wrong.

Name: _____ Date: _____ Period: _____

LAB CORNER • Collecting a precipitate

A student mixes two solutions, filters the solid, dries it and finds its mass.



1. Give three reasons the actual yield of a precipitate is usually less than the theoretical yield. Then give one reason a student might get a percent yield greater than 100 %.

_____

_____

2. Methane burns: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$.

a. A student reacts 10.0 g of O_2 with excess CH_4. Calculate the theoretical yield of CO_2.

Work:

Answer (units & sig figs): _____

b. The lab produces 6.50 g of CO_2. Calculate the percent yield.

Work:

Answer (units & sig figs): _____

3. $CaCl_2\ (aq) + 2AgNO_3\ (aq) \rightarrow Ca(NO_3)_2\ (aq) + 2AgCl\ (s)$

a. A student reacts 15.0 g of $CaCl_2$ with 10. g of $AgNO_3$. Find the limiting reactant and the theoretical yield of AgCl.

Work:

Answer (units & sig figs): _____

b. 4.50 g of AgCl are obtained. Find the percent yield.

Work:

Answer (units & sig figs): _____

4. $AlCl_3 + 3NaOH \rightarrow Al(OH)_3 + 3NaCl$

a. 15.0 g of $AlCl_3$ react with 20.0 g of NaOH. Find the limiting reactant and the theoretical yield of $Al(OH)_3$.

Work:

Answer (units & sig figs): _____

b. 7.30 g of $Al(OH)_3$ are collected. Find the percent yield.

Work:

Answer (units & sig figs): _____

5. A reaction has a theoretical yield of 25.0 g. A student collects 21.0 g. The percent yield is…

A) 84.0 % B) 119 % C) 16.0 % D) 46.0 g

Name: _____ Date: _____ Period: _____

PRACTICE

Extra Practice — Unit 6

Show every setup with units.

A. Mole ratios and mass-to-mass

1. Use $K_3PO_4 + Al(NO_2)_3 \rightarrow 3KNO_2 + AlPO_4$.

a. How many grams of KNO_2 form from 4.6 mol of K_3PO_4?

Work:

Answer (units & sig figs): _____

b. If 5.8 g of $AlPO_4$ form, how many grams of $Al(NO_2)_3$ were used?

Work:

Answer (units & sig figs): _____

c. How many grams of KNO_2 form from 6.2 g of K_3PO_4?

Work:

Answer (units & sig figs): _____

2. Use $CO + 2H_2 \rightarrow CH_3OH$.

a. If 5.7 g of CO are used, how many grams of H_2 are also used?

Work:

Answer (units & sig figs): _____

b. If 0.89 g of H_2 are used, how many grams of CH_3OH form?

Work:

Answer (units & sig figs): _____

c. If 15.0 L of CO are used at STP, how many liters of H_2 are also used? Why is no molar mass needed?

Work:

Answer (units & sig figs): _____

B. Limiting reactant and percent yield

3. 10.0 g of Al react with 20.0 g of Cl_2: $2Al + 3Cl_2 \rightarrow 2AlCl_3$. Which is limiting, and how many grams of $AlCl_3$ can form? How much of the excess reactant is left?

(mol)	2Al	$3Cl_2$	$2AlCl_3$
Before			
Change			
After			

Work and answers:

4. 6.54 g of Zn react with 0.150 mol of HCl: $Zn + 2HCl \rightarrow ZnCl_2 + H_2$. Find the liters of H_2 at STP that can form and the grams of the excess reactant left.

(mol)	Zn	2HCl	$ZnCl_2$	H_2
Before				
Change				
After				

Work and answers:

5. A student burns 4.86 g of Mg in air: $2Mg + O_2 \rightarrow 2MgO$. The white powder has a mass of 7.60 g.

a. Calculate the theoretical yield of MgO.

Work:

Answer (units & sig figs): _____

b. Calculate the percent yield and suggest a reason it differs from 100 %.

Work:

Answer (units & sig figs): _____

C. Lab data

6. A student mixes 2.12 g of Na_2CO_3 (in water) with excess $CaCl_2$ solution: Na_2CO_3 (aq) + $CaCl_2$ (aq) → $CaCO_3$ (s) + 2NaCl (aq). The precipitate is filtered, dried and weighed. Data: filter paper 0.86 g; filter paper + dry solid 2.76 g.

(a) Actual yield (b) theoretical yield (c) percent yield

D. Think

7. A recipe makes 24 cookies from 2 cups of flour. You have 5 cups of flour but only 3 eggs, and the recipe needs 1 egg per 8 cookies. How many cookies can you make, and what is the limiting ingredient? Relate this to a chemical reaction.

8. Why must you convert to moles before deciding which reactant is limiting?

9. $N_2 + 3H_2 \rightarrow 2NH_3$. If 4 mol N_2 and 9 mol H_2 are mixed, the limiting reactant is…

A) N_2 B) H_2 C) NH_3 D) neither

10. Which quantity cannot be negative or over 100 % in a lab report?

A) percent yield B) actual yield minus theoretical C) mole ratio D) Change in a BCA table

Unit 6 review

11. Write the complete plan (in words) for solving “grams of A → grams of B”. Then say what changes for “liters of gas A → grams of B”.

12. Describe how you find the limiting reactant and how you use it to find the product and the leftover.